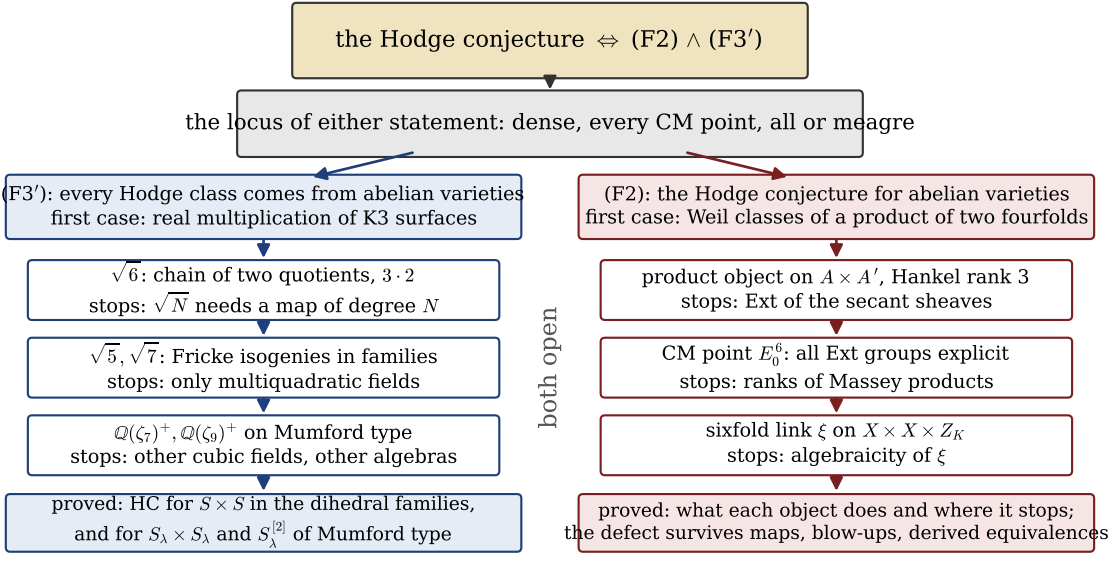
the Hodge conjecture ⇔ (F2) ∧ (F3′)
the locus of either statement: dense, every CM point, all or meagre
(F3′): every Hodge class comes from abelian varieties
first case: real multiplication of K3 surfaces
$\sqrt{6}$: chain of two quotients, $3 \cdot 2$
stops: $\sqrt{N}$ needs a map of degree N
$\sqrt{5}, \sqrt{7}$: Fricke isogenies in families
stops: only multiquadratic fields
$\mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\zeta_9)^+$ on Mumford type
stops: other cubic fields, other algebras
proved: HC for $S \times S$ in the dihedral families,
and for $S_\lambda \times S_\lambda$ and $S_\lambda^{[2]}$ of Mumford type
both open
(F2): the Hodge conjecture for abelian varieties
first case: Weil classes of a product of two fourfolds
product object on $A \times A'$, Hankel rank 3
stops: Ext of the secant sheaves
CM point E_0^6: all Ext groups explicit
stops: ranks of Massey products
sixfold link ξ on $X \times X \times Z_K$
stops: algebraicity of ξ
proved: what each object does and where it stops;
the defect survives maps, blow-ups, derived equivalences